

Dataset 1 Attributes

This document contains a summary of the coursework data that you may find helpful. This document is entirely descriptive to help you understand the dataset better and does not necessarily need to be used or referenced anywhere in the coursework report.

Data description:

This file contains 1000 observations of metal parts that have been destructively tested to estimate their true lifespan for if the part was used for its intended purpose. These lifespan measurements have been collected with processing parameters used when casting the parts, along with observations of the state of the microstructure and observable defects on the completed part.

The COMP1801 coursework Dataset 1 has the following attributes:

- A single .csv file
- 1001 lines
- Line 1 of data provides column headings

Column Headings:

- Lifespan: Continuous Data. Measured lifespan of a metal part (Hours).
- partType: Categorical Data. Different metal parts being tested. (Blade, Nozzle, Valve, Block).
- microstructure: Categorical Data. Observed microstructure for the cast part - equiaxed, columnar or single crystal ('equiGrain', 'singleGrain', 'colGrain').
- coolingRate: Integer Data. Speed of liquid metal cooling, controlled with external heaters. (K/s).
- quenchTime: Continuous Data. Time completed metal part was immersed in water (seconds).
- forgeTime: Continuous Data. Time completed metal part was worked with a automated hammer (seconds).
- HeatTreatTime: Continuous Data. Time completed metal part was given heat treatment (minutes).
- Nickel%: Continuous Data. Percentage of Nickel in the metal alloy.
- Iron%: Continuous Data. Percentage of Iron in the metal alloy.
- Cobalt%: Continuous Data. Percentage of Cobalt in the metal alloy.
- Chromium%: Continuous Data. Percentage of Chromium in the metal alloy.
- smallDefects: Integer Data. Number of small radius defects observed.
- largeDefects: Integer Data. Number of large radius defects observed.
- sliverDefects: Integer Data. Number of elongated defects observed.
- seedLocation: Categorical Data. Notes whether growth begins at top or bottom of mould (Top, Bottom).
- castType: Categorical Data. Type of casting method used (Die, Investment, Continuous).